

- ① Find the general solution of the differential equation:-

$$\cos x \frac{dy}{dx} + y = \sin x$$

- ② From a pack of cards, one card is lost. From the remaining cards of the pack, two cards are drawn and are found to be both diamonds. Find the probability of the lost card being diamond.

- ③ Find the values of p for which the lines are at right angles

$$\frac{1-x}{3} = \frac{7y-14}{p} = \frac{z-3}{2} \text{ and } \frac{7-7x}{3p} = \frac{y-5}{1} = \frac{6-z}{5}$$

- ④ Find the shortest distance between the lines

$$\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1} \text{ and } \frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$$

- ⑤ If $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}| = 4$ and $|\vec{b}| = 5$, and the angle between them is 30° , then $|\vec{a} \times \vec{b}| = ?$

- ⑥ Find the area of the region bounded by the curve $xy^2 = x$ and the lines $x=1$ and $x=4$ and the x -axis?

⑦ Integrate the following :-

a) $\int x \sin^2 x \, dx$

b) $\int e^x \cos x \, dx$

c) $\int e^x (\sec x) (1 + \tan x) \, dx$

d) $\int \frac{4x+1}{\sqrt{2x^2+x-3}} \, dx$

e) $\int \frac{5x}{(x+1)(x^2-4)} \, dx$

⑧ $\int \frac{1}{e^x - 1} \, dx$

g) $\int \frac{1}{9 \sin^2 x + 4 \cos^2 x} \, dx$

h) $\int \sin 4x \cdot \sin 8x \, dx$

i) $\int \sqrt{1-4x-x^2} \, dx$

j) $\int \sin^2(2x+1) \, dx$